



PSTAT 5A: Homework 01

Summer Session B 2025, with Annie Adams

1. Consider the list of numbers $X = \{-3, -1, 0, 0.4, 0.7, 3.9, 6\}$.

(a) Compute the mean of X .

$$\bar{x} = \frac{1}{7} [(-3) + (-1) + (0) + (0.4) + (0.7) + (3.9) + (6)] = \frac{1}{7} \cdot 7 = 1$$

(b) Compute $\text{median}(X)$, the median of X .

$$\{-3, -1, 0, 0.4, 0.7, 3.9, 6\} \Rightarrow \text{median}(X) = 0.4$$

(c) Compute the standard deviation of X .

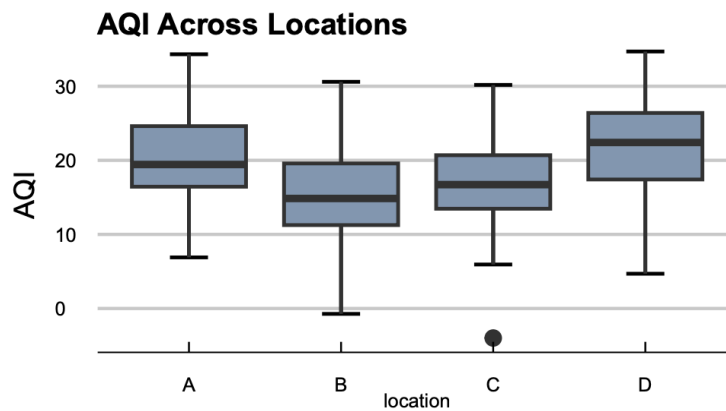
$$\begin{aligned} s_x^2 &= \frac{1}{7-1} \sum_{i=1}^7 (x_i - \bar{x})^2 \\ &= \frac{1}{6} [(-3-1)^2 + (-1-1)^2 + (0-1)^2 + (0.4-1)^2 + (0.7-1)^2 + (3.9-1)^2 + (6-1)^2] \\ &= \frac{1}{6} \cdot \frac{2743}{50} = \frac{2743}{300} \end{aligned}$$

$$\text{Therefore, the standard deviation is } \sqrt{\frac{2743}{300}} = \frac{\sqrt{8229}}{30} \approx 3.023795$$

(d) Compute the range of X .

$$\text{range}(X) = \max\{X\} - \min\{X\} = 6 - (-3) = 9$$

2. A researcher has collected measurements on the AQI (Air Quality Index) at several locations in Santa Barbara. She has labeled her locations A through D, and collected 100 AQI measurements from each location. The results of her study are displayed graphically below:



(a) What is the (approximate) median AQI at location A?

We know that, on a boxplot, the median is depicted by the dark bar in the middle of the box. As such, we see that the median AQI at location A is around 20.

(b) Approximately what percent of AQI readings at Location D were less than 16?

We know that, on a boxplot, the first quartile is depicted by the bottom of the box. This means that the first quartile of AQI readings at location D is around 16, meaning around 25% of AQI readings at Location D were less than 16.

(c) Are there any outliers at any of the locations?

Outliers are depicted by points/circles outside of the reach of the whiskers. Only Location C has such points, meaning only Location C had outliers.

(d) From the plot, does it appear that the different locations have different AQI readings? Explain briefly.

It does appear that the average AQI readings in the different locations are different. Specifically, it seems that Locations B and C have slightly lower average AQI readings than Locations A and D.

3. A recent survey revealed that 40% of UCSB students like Bagel Cafe, 70% like Yetz's Bagels, and 30% like both Bagel Cafe and Yetz's Bagels. A UCSB student is selected at random.

Let X denote the event "student likes Bagel Cafe" and P denote the event "student likes Yetz's Bagels". From the information provided, we have

$$\mathbb{P}(X) = 0.4$$

$$\mathbb{P}(P) = 0.7$$

$$\mathbb{P}(X \cap P) = 0.3$$

- (a) What is the probability that this student likes Bagel Cafe or Yetz's Bagels (*or both*)?

We seek $\mathbb{P}(X \cup P)$, which can be computed using the Addition Rule:

$$\mathbb{P}(X \cup P) = \mathbb{P}(X) + \mathbb{P}(P) - \mathbb{P}(X \cap P)$$

$$= (0.4) + (0.7) - (0.3)$$

$$= 0.8$$

(b) What is the probability that this student likes Bagel Cafe but not Yetz's Bagels?

$$\mathbb{P}(P \cap X^c) = \mathbb{P}(P) - \mathbb{P}(X \cap P)$$

$$= 0.7 - 0.3$$

$$= 0.4$$

(c) What is the probability that this student likes neither Bagel Cafe nor Yetz's Bagels?

We seek $\mathbb{P}(X^c \cap P^c)$. The Complement Rule tells us that:

$$\mathbb{P}(X^c \cap P^c) = 1 - \mathbb{P}[(X^c \cap P^c)^c]$$

DeMorgan's Laws tell us that $(X^c \cap P^c)^c = X \cup P$, meaning, putting everything together:

$$\mathbb{P}(X^c \cap P^c) = 1 - \mathbb{P}(X \cup P)$$

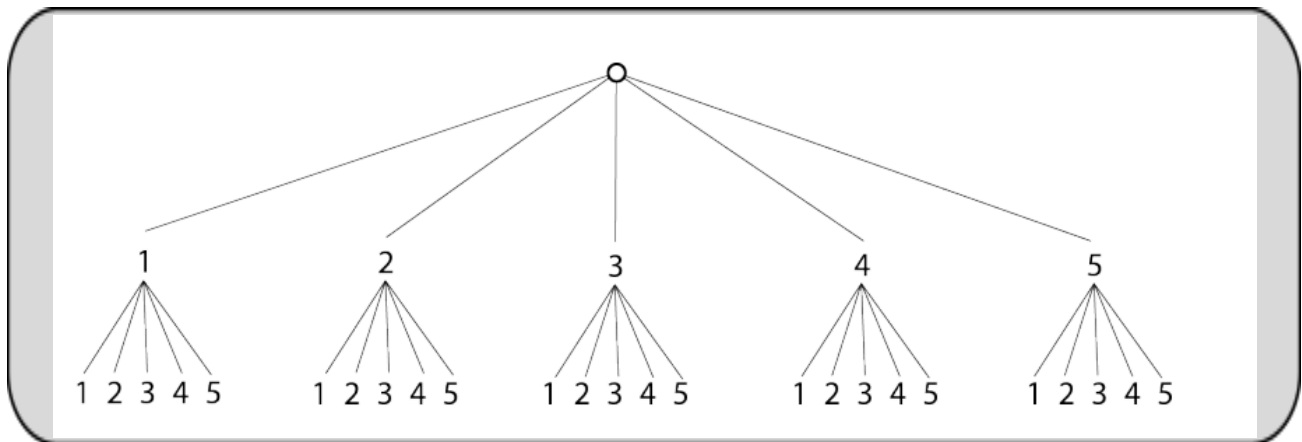
$$= 1 - (0.8)$$

4. Two numbers are to be selected at random and with replacement from the set $\{1, 2, 3, 4, 5\}$.

(a) Express the outcome space for this experiment using a table.

	1	2	3	4	5
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

(b) Express the outcome space for this experiment using a tree.



(c) How many elements are in the outcome space?

Counting either the elements in our table from part (a) or the number of leaves in our tree from part (b), we see that the total number of outcomes in Ω is 25.

(d) For each of the following events, write down the mathematical formulation of the event described (i.e. list out the outcomes comprised in each event):

i. A = the first number selected was 3

$$A = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5)\}$$

ii. B = the second number selected was even

$$B = \{(1, 2), (2, 2), (3, 2), (4, 2), (5, 2), (1, 4), (2, 4), (3, 4), (4, 4), (5, 4)\}$$

iii. C = the first number selected was 3 and the second number selected was even.

$$C = A \cap B = \{(3, 2), (3, 4)\}$$

iv. D = the first number selected was odd, and the first number selected was even.

There are no outcomes in which the first number is simultaneously even and odd; hence $D = \emptyset$.

v. E = the first number was strictly greater than the second.

	1	2	3	4	5
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

(e) Are we justified in using the Classical Approach to Probability in this problem? Why or why not?

We are justified in using the Classical Approach, because the numbers are stated to be selected at random.

(f) Use the Classical Approach to Probability to compute the probabilities of the events listed out in part (d) above.

For each probability, we simply count the number of elements in the event and divide by the total number of outcomes in Ω (which we already computed to be 25).

(i) $\mathbb{P}(A) = \frac{1}{5} = 0.2 = 20\%$

(ii) $\mathbb{P}(B) = \frac{2}{5} = 0.4 = 40\%$

(iii) $\mathbb{P}(C) = \frac{2}{25} = 0.08 = 8\%$

(iv) $\mathbb{P}(D) = 0$

(v) $\mathbb{P}(E) = \frac{2}{5} = 0.4 = 40\%$